Himaayah Schools
ACADEMY

GUIDE

وَجَعَلْنَا مِنَ الْمَاءِ كُلَّ شَيْءٍ حَيٍّ

"And We made from water every living thing." (al-Anbiyāʾ 21:30)

JAMB Biology

A UTME study guide: the syllabus, key facts, genetics made simple, worked examples and a practice test.

بِسْمِ اللَّهِ الرَّحْمٰنِ الرَّحِيمِ

Before you begin

UTME Biology is a wide subject, but most questions test clear facts and understanding: what a structure does, how a process works, and how organisms are related. Genetics and ecology bring a few short calculations. This guide gives you the syllabus in five areas, the key facts to know, genetics step by step, nine worked examples, common traps, an 8-week plan and a practice test.

The UTME at a glance	
Subjects	Use of English and three subjects for your course. Biology is required for many science, medical and engineering courses.
Questions	Use of English 60; each other subject 40; 180 questions in all, on computer (CBT).
Time	About two hours for all four papers, so roughly 40 seconds a question. You move freely between subjects.
Marking	Each subject is scored out of 100, for a total out of 400. There is no negative marking: never leave a question blank.

Check the official details

Exam rules and syllabuses can change from year to year. Always check the current JAMB brochure and the official Biology syllabus on the JAMB website or e-facility portal, and follow them where they differ from this guide.

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Part 1 The syllabus at a glance

Study from the **JAMB syllabus**, not only from your school notes: every UTME question comes from a topic on it. The topics fall into these areas:

Area	Topics
Variety of organisms	Characteristics of living things; the cell and its organelles; levels of organisation; classification: kingdoms Monera, Protista, Fungi, Plantae and Animalia, with examples; adaptations; viruses
Form and functions	Nutrition in plants (photosynthesis, mineral requirements) and animals (food classes, digestion, enzymes); transport; respiration; excretion; support and movement; reproduction; growth; coordination: nervous system, sense organs, hormones in animals and plants
Ecology	Ecological factors and habitats; food chains, webs and pyramids; energy flow; nutrient cycles (carbon, nitrogen, water); population studies; symbiosis; conservation; pollution; pests and diseases
Heredity and variation	Variation; Mendel's laws; monohybrid crosses; blood groups; sex determination and sex-linked traits; the sickle-cell genotype; applications of genetics in agriculture and medicine
Evolution	Theories of evolution (Lamarck, Darwin); evidence for evolution; adaptation and natural selection

Form and functions is the largest area. Do not neglect ecology and genetics, which appear every year and are among the easiest marks once understood.

Part 2 Key facts at a glance

The cell

Organelle	Function	Found in
Nucleus	Controls the cell; contains the chromosomes (DNA)	Plant and animal cells
Mitochondrion	Aerobic respiration: releases energy (ATP)	Plant and animal cells
Chloroplast	Photosynthesis	Plant cells only
Cell wall	Cellulose; gives shape and support	Plant cells (not animal cells)
Cell membrane	Controls what enters and leaves	All cells
Ribosome	Protein synthesis	All cells
Vacuole	Large and permanent in plants; stores cell sap	Mainly plant cells

Processes and equations

Process	Summary	Note
Photosynthesis	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (light, chlorophyll)	Oxygen comes from water
Aerobic respiration	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy	In mitochondria
Anaerobic: yeast	Glucose → ethanol + CO_2 + a little energy	Fermentation
Anaerobic: muscle	Glucose → lactic acid + a little energy	Causes muscle fatigue

Digestion and food tests

Item	Detail	Result or product
Salivary amylase (ptyalin)	Mouth: starch	Maltose
Pepsin	Stomach (acid): protein	Polypeptides
Trypsin, lipase, amylase	Small intestine (pancreatic juice)	Peptides; fatty acids and glycerol; maltose
Bile	From the liver, stored in the gall bladder: emulsifies fats	No enzyme
Starch test	Iodine solution	Blue-black
Reducing sugar test	Benedict's or Fehling's solution, heated	Brick-red
Protein test	Biuret test (or Millon's reagent)	Violet (or brick-red with Millon's)
Fat test	Sudan III, or grease spot on paper	Red layer; translucent spot

Hormones

Hormone	Gland	Function
Insulin	Pancreas (islets of Langerhans)	Lowers blood glucose
Adrenaline	Adrenal gland	Prepares the body for "fight or flight"
Thyroxine	Thyroid	Controls metabolic rate; lack causes goitre (with iodine deficiency)
ADH	Pituitary	Increases water reabsorption in the kidney
FSH, LH	Pituitary	Control the menstrual cycle and ovulation
Auxin	Shoot and root tips (plants)	Growth; phototropism and geotropism

The nitrogen cycle

Organism	Role	
Rhizobium	Nitrogen fixation in root nodules of legumes	$N_2 \rightarrow$ compounds
Nitrosomonas	Nitrification	Ammonium → nitrite
Nitrobacter	Nitrification	Nitrite → nitrate
Denitrifying bacteria	Denitrification	Nitrate → N_2
Lightning	Fixes nitrogen into oxides, washed into the soil by rain	

Genetics in one line: dominant alleles are written in capitals (T), recessive in small letters (t). A homozygous organism has two identical alleles (TT or tt); a heterozygous one has two different alleles (Tt). The genotype is the alleles; the phenotype is what you see.

Part 3 Worked examples

Example 1. Two people with genotype AS have children. What fraction of the children are expected to be SS (sickle-cell disease)?

→ Gametes: A, S from each parent

→ Cross: AA, AS, AS, SS

→ Ratio 1 AA : 2 AS : 1 SS

Answer: 1/4 (25%) SS; 1/2 AS carriers; 1/4 AA

Example 2. A man of blood group A (genotype I^AI^O) marries a woman of group B (I^BI^O). What blood groups can their children have?

→ Gametes: I^A, I^O and I^B, I^O

→ Children: I^AI^B (AB), I^AI^O (A), I^BI^O (B), I^OI^O (O)

Answer: AB, A, B or O, each with a chance of 1/4

Example 3. In peas, tall (T) is dominant to short (t). A heterozygous tall plant is crossed with a short plant. Give the ratio of the offspring.

→ Tt × tt; gametes T, t and t

→ Offspring: Tt, Tt, tt, tt

Answer: 1 tall : 1 short (a test cross)

Example 4. A woman who is a carrier of colour blindness (X^cX) marries a man with normal vision (XY). What are the chances for their children?

→ Gametes: X^c, X and X, Y

→ Children: X^cX (carrier daughter), XX (normal daughter), X^cY (colour-blind son), XY (normal son)

Answer: Half the sons colour-blind; half the daughters carriers; no colour-blind daughters

Example 5. Two tall plants (Tt × Tt) are crossed. What is the ratio of the phenotypes?

→ Offspring: TT, Tt, Tt, tt

→ 3 have T (tall), 1 is tt (short)

Answer: 3 tall : 1 short

Example 6. In a food chain, producers fix 10 000 kJ of energy. If about 10% passes to each level, how much reaches the secondary consumers?

→ Primary consumers: 10% of 10 000 = 1000 kJ

→ Secondary consumers: 10% of 1000 = 100 kJ

Answer: About 100 kJ

Example 7. A cell 0.05 mm long appears 20 mm long in a drawing. What is the magnification?

→ Magnification = image size ÷ actual size = 20 ÷ 0.05

Answer: × 400

Example 8. Which part of the nephron does ultrafiltration take place in?

→ Blood at high pressure in the glomerulus is filtered into Bowman's capsule

→ Selective reabsorption follows in the tubules

Answer: The glomerulus and Bowman's capsule

Example 9. Classify: a house fly, a spider, a millipede and a crab.

→ Insects: 6 legs, 3 body parts, often wings: house fly

→ Arachnids: 8 legs, 2 body parts: spider

→ Myriapods: many legs: millipede; Crustaceans: two pairs of antennae, hard shell: crab

Answer: All are arthropods, in four different classes

Part 4 Traps that cost marks

Trap	What to do
Confusing genotype and phenotype	Genotype is the alleles (Tt); phenotype is the visible trait (tall). Ratios of each can differ.
Thinking plants do not respire	Plants respire all the time, day and night; they photosynthesise only in light.
Arteries and veins	Arteries carry blood away from the heart (all oxygenated except the pulmonary artery); veins carry blood to the heart (all deoxygenated except the pulmonary vein).
Mixing up the enzymes	Learn enzyme, where it works, its substrate and its product as one set.
Pyramid of numbers vs energy	A pyramid of numbers can be inverted (one tree, many insects); a pyramid of energy never is.
Sex-linked traits in girls	A daughter shows a sex-linked recessive trait only if she inherits it from both parents.
Monocots and dicots	Monocots: one cotyledon, parallel veins, fibrous roots (maize). Dicots: two cotyledons, net veins, tap root (beans).
Malaria	Caused by Plasmodium (a protist), carried by the female Anopheles mosquito. The mosquito is the vector, not the cause.
Not reading "which is NOT"	Underline NOT, EXCEPT and LEAST before you answer.

Part 5 An 8-week plan

Start at least eight weeks before the exam. Each week: learn the topics, make short notes and solve past questions under time. Keep one day a week for revision.

Week	Focus	Practice
1	Living things, the cell, levels of organisation, classification	Labelled diagrams; classification questions
2	Nutrition in plants and animals; enzymes; food tests	Digestion and food test questions
3	Transport in plants and animals; respiration	Heart, blood and transpiration questions
4	Excretion; support and movement; skeleton and joints	Kidney, skin and skeleton diagrams
5	Reproduction in plants and animals; growth	Flower, fruit and human reproduction questions
6	Coordination: nervous system, sense organs, hormones, plant responses	Eye, ear and hormone questions
7	Ecology; genetics; evolution	Genetic crosses and food chain calculations
8	Revision of weak areas; full timed mock exams	Two full CBT mocks; review every error

Exam-day reminders

Read every option before choosing. Eliminate the clearly wrong ones first.

Do not spend more than a minute on one question: flag it and come back.

Answer every question; there is no penalty for a wrong answer.

Check that your answers are saved before you submit, and keep an eye on the clock for all four subjects.

Test Practice test

Give yourself 10 minutes for these 15 questions, as in the real exam. The answers follow.

1. The organelle where aerobic respiration takes place is the:

a) Ribosome b) Mitochondrion c) Nucleus d) Chloroplast

2. Which of these is found in plant cells but not in animal cells?

a) Cell membrane b) Nucleus c) Cell wall d) Mitochondrion

3. The end product of protein digestion is:

a) Amino acids b) Glucose c) Fatty acids d) Glycerol

4. A food that gives a violet colour with Biuret reagent contains:

a) Starch b) Fat c) Reducing sugar d) Protein

5. The blood vessel that carries oxygenated blood from the lungs to the heart is the:

a) Pulmonary artery b) Aorta c) Pulmonary vein d) Vena cava

6. Which hormone lowers blood sugar?

a) Adrenaline b) Insulin c) Thyroxine d) Glucagon

7. Two carriers (AS × AS) have a child. The chance that the child is AS is:

a) 1/4 b) 1/2 c) 3/4 d) 0

8. A person with blood group O can receive blood only from group:

a) A b) B c) AB d) O

9. Bacteria in the root nodules of legumes are involved in:

a) Nitrogen fixation b) Denitrification c) Photosynthesis d) Decomposition only

10. In a food chain, the organisms with the most energy available are the:

a) Producers b) Primary consumers c) Secondary consumers d) Top carnivores

11. Malaria is caused by:

a) Anopheles b) A virus c) Plasmodium d) Trypanosoma

12. Which of these is an insect?

a) Spider b) Grasshopper c) Millipede d) Scorpion

13. The theory of natural selection was proposed by:

a) Lamarck b) Mendel c) Linnaeus d) Darwin

14. The functional unit of the kidney is the:

a) Nephron b) Neuron c) Alveolus d) Villus

15. Water moves from a dilute solution to a concentrated one through a membrane by:

a) Diffusion b) Active transport c) Osmosis d) Transpiration

No.	Answer	No.	Answer
1	b) Mitochondrion	9	a) Nitrogen fixation
2	c) Cell wall	10	a) Producers
3	a) Amino acids	11	c) Plasmodium
4	d) Protein	12	b) Grasshopper
5	c) Pulmonary vein	13	d) Darwin
6	b) Insulin	14	a) Nephron
7	b) 1/2	15	c) Osmosis
8	d) O		

Recommended textbooks

- **Modern Biology for Senior Secondary Schools**, S. T. Ramalingam.
- **Essential Biology**, M. C. Michael.
- Other texts listed in the current JAMB Biology syllabus.
- **Past questions**: solve at least the last ten years' UTME Biology questions, timing yourself, and study every one you get wrong.

Prepare with the Himaayah Academy

Live UTME, Post-UTME and O-Level classes with a personal mentor, daily tasks, weekly mock exams, CBT practice and a leaderboard, all online.

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وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ

"And my success is only through Allah." (Hūd 11:88)

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Learn the living world, one system at a time.

This guide from the Himaayah Academy helps UTME candidates prepare for Biology with a clear plan, the facts that come up every year and genetics explained step by step.

- The JAMB Biology syllabus in five areas
- Cells, nutrition, transport and coordination at a glance
- Genetics crosses, including the sickle-cell genotype
- Ecology, classification and food tests
- An 8-week plan and a 15-question practice test

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